Understand the whole position.

Educational draft. Current symptoms, restrictions and tolerated loads have not been confirmed. Review suitability with a qualified clinician or trainer where injury, pain or medical restrictions apply.

Three roles / one complete position

MAIN / RED: intended movement focus.

SECONDARY / BLUE: movement assistants and local joint control.

BASE / GREEN: postural support and stable contact points.

BAND / ORANGE-EDGED CHARCOAL: the resistance line at every level. Colour never specifies load.

Read in order: setup > base and brace > move > return

Inspect the equipment and stable supports first. Warm up with comfortable activity and rehearse unloaded. Brace gently while breathing; avoid forced joint locks. Use only a range you can control.

1. Bench-supported one-arm row
2. Supported single-arm chest press
3. Seated straight-bar pressdown
4. Short-arc shoulder raise
5. Seated brace with normal breathing
6. Supported sit-to-stand
7. Seated single-leg band curl
8. Supported calf raise
9. Supported side leg lift
10. Supported shoulder-blade control
11. Seated neutral-grip curl
12. Supported backward leg lift

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Bench-supported one-arm row

Pull / back / 8-12 each side



START / softly extended arm FINISH / controlled row

MAIN: Latissimus dorsi: draws the upper arm back toward the trunk.

SECONDARY: Rhomboids and middle trapezius control the shoulder blade; rear deltoid and elbow flexors assist the pull.

BASE / SUPPORT: Abdominals and spinal stabilizers resist rotation. Hips, thighs and ankle muscles steady both feet; the other arm supports the torso.

01 / Check the setup

Use a stable bench and a secure front rack anchor near the working hand’s starting height. Clip one handle to one continuous pull line; keep the recoil path clear.

02 / Set the base and brace

Plant both feet in a comfortable stagger. Support the opposite palm under its shoulder. Incline from the hips; keep neck and trunk aligned, knees soft.

03 / Move with control

Gently brace while breathing. Exhale and guide the working elbow toward the lower ribs; keep the shoulder away from the ear.

04 / Limit the range and return

Stop before the torso twists, the shoulder rolls forward or the elbow has to travel far behind the body. Return slowly to a softly extended arm.

WATCH FOR: Both feet and the support hand stay planted. Shorten the pull or reduce resistance if you shrug, jerk or rotate. EASIER: Rehearse unloaded, use less tension, or use a more upright supported row if the hinge is uncomfortable.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: ACE - Single-arm Row; ACE - Back exercise research; ACE - Resistance-band workout builder. Full source links: project100.fit/downloads/kinetics/index.html

Supported single-arm chest press

Push / chest / 8-12 each side

MAIN: Pectoralis major: moves the upper arm forward and across the chest.

SECONDARY: Triceps extend the elbow; anterior deltoid assists. Scapular muscles guide the shoulder blade.

BASE / SUPPORT: Trunk muscles resist rotation; the chair and planted feet stabilize the pelvis and legs.

01 / Check the setup

Use a stable non-rolling chair and a secure anchor behind you near chest height. Keep the band beside the arm, away from the neck and face.

02 / Set the base and brace

Sit tall with both feet planted. Start with the handle beside the chest, neutral wrist and elbow below shoulder height. Let the shoulder blade move naturally.

03 / Move with control

Breathe out and press forward smoothly while keeping ribs and pelvis steady. Do not lean or twist to finish.

04 / Limit the range and return

Finish before the elbow locks; return only as far as the shoulder remains comfortable. Avoid a deep forced stretch behind the torso.

WATCH FOR: If the ribs flare, shoulder pinches or trunk turns, shorten the range and lower the load. EASIER: Use an unloaded press rehearsal or a comfortable wall press-up instead.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: ACE - Resistance-band workout builder; NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html

Seated straight-bar pressdown

Elbow extension / triceps / 8-12

MAIN: Triceps brachii: straightens the elbow.

SECONDARY: Forearm muscles steady the grip and wrist; shoulder muscles hold the upper arms beside the torso.

BASE / SUPPORT: Trunk muscles hold a quiet seated posture; both feet and the chair provide support.

01 / Check the setup

Use one secure high rack anchor, one rated pull line and a straight-bar attachment. Position a stable chair so the line stays in front of the body.

02 / Set the base and brace

Plant both feet; sit tall. Keep upper arms beside the ribs and wrists in line with forearms. Choose a grip you can hold without strain.

03 / Move with control

Exhale and extend the elbows without pushing the shoulders down with body weight. Keep the upper arms nearly still.

04 / Limit the range and return

Stop short of snapping the elbows straight. Return gradually into comfortable elbow flexion without letting the bar rise toward the face.

WATCH FOR: Avoid a rocking torso, bent wrists or moving the whole upper arm. Reduce load if the elbows or grip lose control. EASIER: Rehearse the elbow motion unloaded or use a lighter single-handle variation matched to current tolerance.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: ACE - Resistance-band workout builder. Full source links: project100.fit/downloads/kinetics/index.html

Short-arc shoulder raise

Shoulder control / 6-10 easy repetitions

MAIN: Middle deltoid contributes to lifting the arm out to the side.

SECONDARY: Supraspinatus assists the lift; rotator-cuff and scapular muscles coordinate the shoulder.

BASE / SUPPORT: Trunk muscles and planted feet keep the torso quiet; a chair supports the seated base.

01 / Check the setup

Begin without a band. Use a stable chair with room at both sides. This is a comfortable movement rehearsal, not a heavy raise.

02 / Set the base and brace

Sit tall, feet planted, elbows softly bent and hands relaxed. Start with arms beside the body.

03 / Move with control

Lift in a slightly forward plane with thumbs comfortably forward or up. Breathe normally and keep the neck relaxed.

04 / Limit the range and return

Use a small pain-free arc below shoulder height; stop sooner if the shoulder hikes, pinches or the torso leans.

WATCH FOR: Do not force the arms overhead or use a shrug to gain height. A smaller controlled arc is the aim. EASIER: Move one arm at a time through a shorter comfortable arc, or omit if symptoms occur.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: AAOS - Shoulder Conditioning Program; ACE - Resistance-band workout builder. Full source links: project100.fit/downloads/kinetics/index.html

Seated brace with normal breathing

Trunk stability / 3 relaxed breaths per hold; 3 holds

MAIN: Abdominal wall: develops gentle trunk stiffness while breathing.

SECONDARY: Deep spinal stabilizers cooperate with the abdominal wall; breathing muscles continue their normal work.

BASE / SUPPORT: Chair contact and planted feet steady the pelvis. Hip and leg muscles provide light postural support.

01 / Check the setup

Use a stable chair. No band is required; this is a low-effort control drill.

02 / Set the base and brace

Sit comfortably tall with ribs over pelvis and both feet on the floor. Keep shoulders relaxed and avoid arching the back.

03 / Move with control

Gently firm the midsection as if preparing for a light nudge. Take slow normal breaths without bearing down.

04 / Limit the range and return

Hold only while breathing and posture stay easy. Release fully between attempts; no trunk movement is needed.

WATCH FOR: Do not suck the stomach in forcefully, brace maximally or hold the breath.
EASIER: Use a lighter contraction for fewer breaths with back support if comfortable.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: AAOS - Spine Conditioning Program. Full source links: project100.fit/downloads/kinetics/index.html

Supported sit-to-stand

Knee / hip extension / 6-10

MAIN: Quadriceps and gluteus maximus raise the body from the chair.

SECONDARY: Hamstrings and hip muscles coordinate the hip and knee movement.

BASE / SUPPORT: Trunk and ankle muscles help balance the body over the feet; the chair marks the return position.

01 / Check the setup

Place a stable chair against a wall. Choose a seat height you can rise from without dropping or straining. Start without bands.

02 / Set the base and brace

Sit near the front with feet about hip-width apart, heels planted and knees following the toes. Use armrests for assistance if needed.

03 / Move with control

Lean forward from the hips enough to bring weight over the feet, then stand smoothly. Breathe out during the effort.

04 / Limit the range and return

Finish upright with soft knees. Bend hips and knees to return under control; touch down rather than falling into the chair.

WATCH FOR: Avoid knees collapsing inward, breath-holding or pushing through pain. Use a higher seat or hand assistance. EASIER: Use a higher chair or more hand support; keep the same controlled return.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html

Seated single-leg band curl

Knee flexion / 8-12 each side

MAIN: Hamstrings bend the knee, drawing the heel back.

SECONDARY: Gastrocnemius can assist knee flexion; hip muscles help keep the thigh aligned.

BASE / SUPPORT: Chair, opposite planted foot and trunk muscles steady the pelvis. The working thigh stays supported.

01 / Check the setup

Use a stable chair and a low anchor in front. Attach a manufacturer-approved ankle cuff to a light band, clear of chair legs and the other foot.

02 / Set the base and brace

Sit tall with the working thigh supported. Start with that heel forward and the knee softly extended; keep the ankle comfortable.

03 / Move with control

Exhale and slide or draw the heel back by bending the knee. Keep the thigh and pelvis quiet rather than pulling with the hip.

04 / Limit the range and return

Use only a comfortable arc with enough clearance under the chair. Return slowly; do not allow a sudden recoil or locked knee.

WATCH FOR: If the chair shifts, cuff slips, knee hurts or pelvis rocks, stop and reset; use less tension. EASIER: Rehearse the heel slide without a band. Do not add resistance until the cuff and anchor are appropriate.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: ACE - Resistance-band workout builder; NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html

Supported calf raise

Ankle plantarflexion / 8-12

MAIN: Gastrocnemius and soleus lift the heels.

SECONDARY: Other lower-leg and foot muscles control ankle alignment.

BASE / SUPPORT: Hip and trunk muscles keep the body upright; hands on a fixed support assist balance.

01 / Check the setup

Use a fixed counter or stable chair against a wall. Begin on a flat floor without extra resistance.

02 / Set the base and brace

Stand with feet about hip-width apart, weight shared between them and knees soft. Hold support lightly but securely.

03 / Move with control

Rise slowly onto the forefeet while keeping pressure across the ball of each foot; breathe normally.

04 / Limit the range and return

Lift only as high as you can stay balanced. Lower heels gently to the floor without bouncing.

WATCH FOR: Do not roll ankles outward, lean heavily forward or grip with the toes to force more height. EASIER: Use a smaller lift or do seated heel raises if standing support is uncomfortable.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html

Supported side leg lift

Hip abduction / 6-10 each side

MAIN: Gluteus medius and minimus lift the thigh away from the midline.

SECONDARY: Tensor fasciae latae assists; other hip muscles coordinate the leg position.

BASE / SUPPORT: The stance-side hip, trunk and ankle muscles steady the pelvis; hands assist balance on a fixed support.

01 / Check the setup

Start without a band beside a fixed counter or stable chair. Leave room to move one leg sideways.

02 / Set the base and brace

Stand tall with the pelvis level and stance knee soft. Keep toes facing mostly forward.

03 / Move with control

Lift the free leg a short distance to the side while breathing normally, then lower with control.

04 / Limit the range and return

Stop before the pelvis tilts, toes rotate upward or the trunk leans away. Keep the motion small and deliberate.

WATCH FOR: The movement comes from the hip, not a swinging leg or sideways bend of the waist. EASIER: Use a smaller lift with more hand support; omit if the stance hip or knee is painful.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html

Supported shoulder-blade control

Scapular control / 6-10 easy repetitions

MAIN: Middle trapezius and rhomboids gently retract the shoulder blades.

SECONDARY: Other scapular and rotator-cuff muscles coordinate the shoulders; this is not a maximal squeeze.

BASE / SUPPORT: Chair, feet and light trunk control maintain an upright seated position.

01 / Check the setup

Sit in a stable chair without a band. Rest hands comfortably on thighs.

02 / Set the base and brace

Keep neck long, ribs settled and shoulders relaxed. Avoid starting in an exaggerated chest-out posture.

03 / Move with control

Gently guide the shoulder blades slightly toward one another, then release. Breathe throughout.

04 / Limit the range and return

Use a small comfortable movement; stop before shrugging, arching or pressing shoulders forcefully down.

WATCH FOR: Do not pinch hard, pull the head backward or add load to compensate for poor control. EASIER: Use a smaller motion or just practice relaxed posture and breathing.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: AAOS - Shoulder Conditioning Program; AAOS - Spine Conditioning Program. Full source links: project100.fit/downloads/kinetics/index.html

Seated neutral-grip curl

Elbow flexion / 8-12 each side

MAIN: Brachialis, biceps brachii and brachioradialis bend the elbow.

SECONDARY: Forearm muscles hold a neutral grip; shoulder muscles steady the upper arm.

BASE / SUPPORT: Chair, feet and trunk control keep the pelvis and torso still.

01 / Check the setup

Use one light handle line from a secure low anchor in front, clear of the feet. A light handheld weight is an alternative.

02 / Set the base and brace

Sit tall with both feet planted. Keep the working upper arm beside the body, thumb up and wrist straight.

03 / Move with control

Breathe out and bend the elbow without swinging the upper arm forward or leaning back.

04 / Limit the range and return

Curl through a comfortable arc, then lower slowly to a softly extended elbow. Keep the band away from the face.

WATCH FOR: Avoid wrist bending, shoulder hiking or a tightening painful grip. Reduce resistance if control changes. EASIER: Rehearse unloaded, use less resistance or choose a light weight you can grip comfortably.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: NHS - Strength exercises; ACE - Resistance-band workout builder. Full source links: project100.fit/downloads/kinetics/index.html

Supported backward leg lift

Hip extension / 6-10 each side

MAIN: Gluteus maximus moves the thigh backward.

SECONDARY: Hamstrings assist hip extension.

BASE / SUPPORT: Stance-leg hip and ankle muscles and trunk control keep the pelvis level; hands assist balance on a fixed support.

01 / Check the setup

Begin without bands, holding a fixed counter or stable chair against a wall.

02 / Set the base and brace

Stand upright with feet under hips and knees soft. Keep ribs over the pelvis and toes facing forward.

03 / Move with control

Move one leg slightly backward from the hip while breathing normally, then return gently.

04 / Limit the range and return

Stop before the lower back arches, pelvis tips forward or the body leans. A small arc is enough.

WATCH FOR: Do not lift higher by bending the spine or swinging the leg. Keep the stance foot planted. EASIER: Use a smaller movement with more hand support, or omit if balance or symptoms limit control.

Soft joints, normal breathing. Colours describe exercise roles, not measured activation. Hidden muscles are named in text. No fixed resistance is prescribed.

Stop for pain, new numbness or weakness, dizziness, loss of grip, moving supports or damaged bands. Keep the recoil path clear. Follow manufacturer limits and individual clinical restrictions.

Sources: NHS - Strength exercises. Full source links: project100.fit/downloads/kinetics/index.html